

Transcriptome Assignment

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Load required packages (you might have to figure out how to install some of these first...)

```
library(ballgown)
library(RColorBrewer)
library(genefilter)
library(dplyr)
library(devtools)
```

This code is creating a data frame with two columns, id and stage.

```
pheno_data<-data.frame(ids = c("plank01", "plank02", "biofilm01", "biofilm02"),
                        stage = c("planktonic", "planktonic", "biofilm", "biofilm"))
```

create Ballgown object and check transcript number

```
samples.c <- paste('ballgown', pheno_data$ids, sep = '/')
bg <- ballgown(samples = samples.c, meas='all', pData = pheno_data)
bg
```

```
## ballgown instance with 5737 transcripts and 4 samples
```

This code is filtering a ballgown object to keep only the genes that show sufficient variability across samples.

```
bg_filt = subset(bg, "rowVars(texpr(bg)) >1", genomesubset=TRUE)
bg_filt
```

```
## ballgown instance with 5163 transcripts and 4 samples
```

create a table of transcripts

```
results_transcripts<- stattest(bg_filt, feature = "transcript", covariate = "stage",
getFC = TRUE, meas = "FPKM")
results_transcripts<-data.frame(geneNames=geneNames(bg_filt),
transcriptNames=transcriptNames(bg_filt), results_transcripts)
```

I chose the transcript gene-PA0001.

```
results_transcripts[results_transcripts$transcriptNames == "gene-PA0001", ]

##   geneNames transcriptNames   feature id      fc      pval      qval
## 1      dnaA      gene-PA0001 transcript  1 52.47471 0.3048003 0.9471885
```

This transcript is giving information on the gene name, the transcript name, the feature, id, fc, pval, and qval.

This code is doing statistical filtering on my transcript results.

```
sigdiff <- results_transcripts %>% filter(pval<0.05)
dim(sigdiff)

## [1] 207  7
```

organize the table <by what metrics is the table being organized?>

```
o = order(sigdiff[, "pval"], -abs(sigdiff[, "fc"]), decreasing=FALSE)
output = sigdiff[o, c("geneNames", "transcriptNames", "id", "fc", "pval", "qval")]
write.table(output, file="SigDiff.txt", sep="\t", row.names=FALSE, quote=FALSE)
head(output)

##   geneNames transcriptNames   id      fc      pval      qval
## 4091      .      gene-PA3992 4091 9.886091e+01 0.0003032315 0.9471885
## 4958      .      gene-PA4804 4958 3.563696e-04 0.0006661432 0.9471885
## 2745      .      gene-PA2690 2745 5.783390e-02 0.0014192618 0.9471885
## 2896      tpm      gene-PA2832 2896 1.786570e+03 0.0023414834 0.9471885
## 370      .      gene-PA0365  370 3.964652e-07 0.0023906201 0.9471885
## 3129      pelF      gene-PA3059 3129 1.687425e-03 0.0025838457 0.9471885
```

load gene names

```
bg_table = texpr(bg_filt, 'all')
bg_gene_names = unique(bg_table[, 9:10])
```

pull out gene expression data and visualize

```
gene_expression = as.data.frame(gexpr(bg_filt))
head(gene_expression)

##           FPKM.plank01 FPKM.plank02 FPKM.biofilm01 FPKM.biofilm02
## .           1.198359    0.9103059      2.526183      2.685373
## MSTRG.1       405.892761  400.8589780     232.324417    181.932617
## MSTRG.10       89.649139   78.5762100      35.010487     59.757320
## MSTRG.100     116.443428  106.2109530      92.206810     95.322479
## MSTRG.1000     7.833186   5.5019700      15.717344     42.342495
## MSTRG.1001     6.845010   4.7381980      38.199095     89.078876
```

This code is renaming the columns of my gene matrix.

```
colnames(gene_expression) <- c("plank01", "plank02", "biofilm01", "biofilm02")
head(gene_expression)
```

```
##           plank01      plank02  biofilm01  biofilm02
## .           1.198359    0.9103059    2.526183    2.685373
## MSTRG.1      405.892761  400.8589780  232.324417  181.932617
## MSTRG.10     89.649139   78.5762100   35.010487   59.757320
## MSTRG.100    116.443428  106.2109530   92.206810   95.322479
## MSTRG.1000    7.833186   5.5019700   15.717344   42.342495
## MSTRG.1001    6.845010   4.7381980   38.199095   89.078876
```

```
dim(gene_expression)
```

```
## [1] 4592    4
```

load the transcript to gene table and determine the number of transcripts and unique genes

```
transcript_gene_table = indexes(bg)$t2g
head(transcript_gene_table)
```

```
##   t_id   g_id
## 1     1 MSTRG.1
## 2     2 MSTRG.2
## 3     3 MSTRG.3
## 4     4 MSTRG.3
## 5     5 MSTRG.4
## 6     6 MSTRG.5
```

```
length(row.names(transcript_gene_table))
```

```
## [1] 5737
```

```
length(unique(transcript_gene_table[, "g_id"]))
```

```
## [1] 4715
```

plot the number of transcripts per gene

```
counts=table(transcript_gene_table[, "g_id"])
c_one = length(which(counts == 1))
c_more_than_one = length(which(counts > 1))
c_max = max(counts)
hist(counts, breaks=50, col="bisque4", xlab="Transcripts per gene",
main="Distribution of transcript count per gene")
legend_text = c(paste("Genes with one transcript =", c_one),
paste("Genes with more than one transcript =", c_more_than_one),
paste("Max transcripts for single gene = ", c_max))
legend("topright", legend_text, lty=NULL)
```

Distribution of transcript count per gene

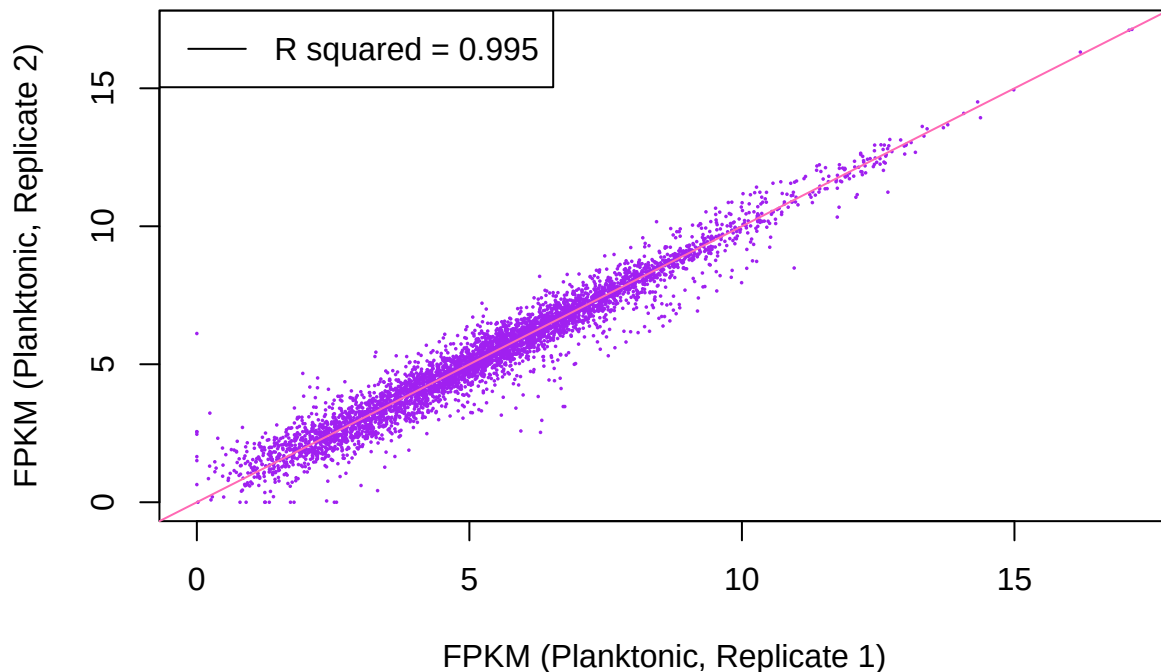


The above graph is plotting the transcript per gene against the frequency at which they appear. There are more genes with only one transcript than anything else.

create a plot of how similar the two replicates are for one another. We have two data sets, I can modify this code in another chunk to create a plot of the other set by looking at both stage data sets (planktonic and biofilm).

```
x = gene_expression[, "plank01"]
y = gene_expression[, "plank02"]
min_nonzero=1
plot(x=log2(x+min_nonzero), y=log2(y+min_nonzero), pch=16, col="purple", cex=0.25,
xlab="FPKM (Planktonic, Replicate 1)", ylab="FPKM (Planktonic, Replicate 2)",
main="Comparison of expression values for a pair of replicates")
abline(a=0,b=1, col = "hotpink")
rs=cor(x,y)^2
legend("topleft", paste("R squared = ", round(rs, digits=3), sep=""), lwd=1, col="black")
```

Comparison of expression values for a pair of replicates

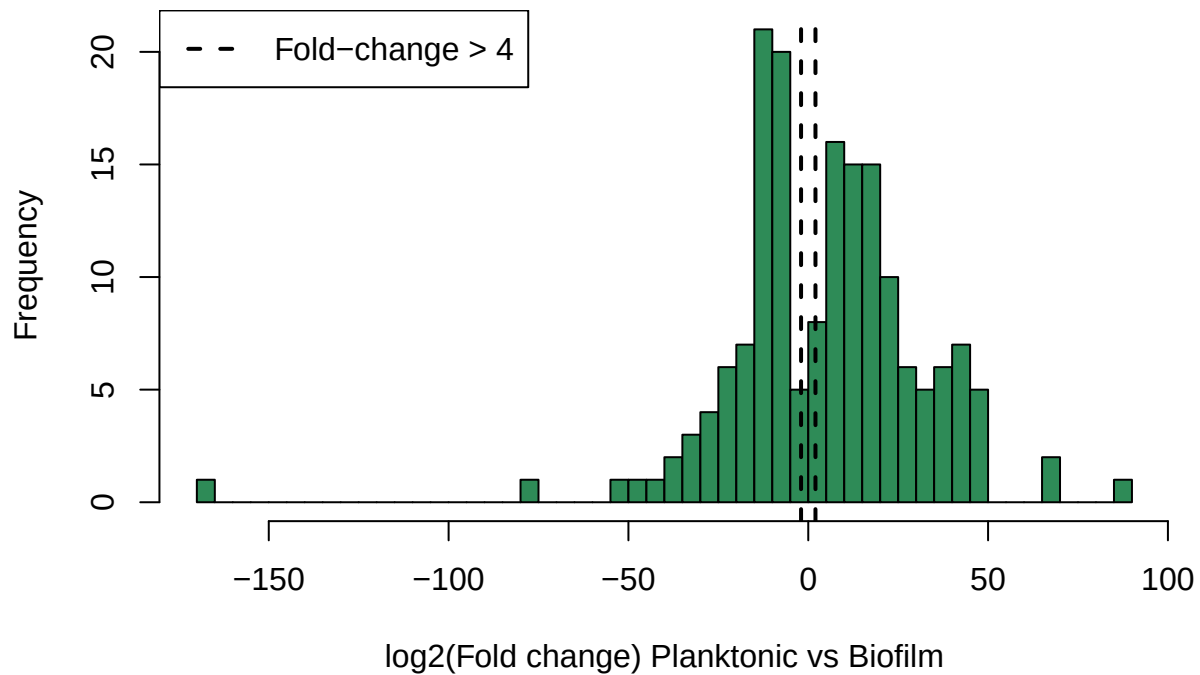


If the two data sets are similar, it shows experimental consistency and that I am using reliable data.

create plot of differential gene expression between the conditions

```
results_genes = statstest(bg_filt, feature="gene", covariate="stage", getFC=TRUE, meas="FPKM")
results_genes = merge(results_genes, bg_gene_names, by.x=c("id"), by.y=c("gene_id"))
sig=which(results_genes$pval<0.05)
results_genes[, "de"] = log2(results_genes[, "fc"])
hist(results_genes[sig, "de"], breaks=50, col="seagreen",
xlab="log2(Fold change) Planktonic vs Biofilm",
main="Distribution of differential expression values")
abline(v=-2, col="black", lwd=2, lty=2)
abline(v=2, col="black", lwd=2, lty=2)
legend("topleft", "Fold-change > 4", lwd=2, lty=2)
```

Distribution of differential expression values

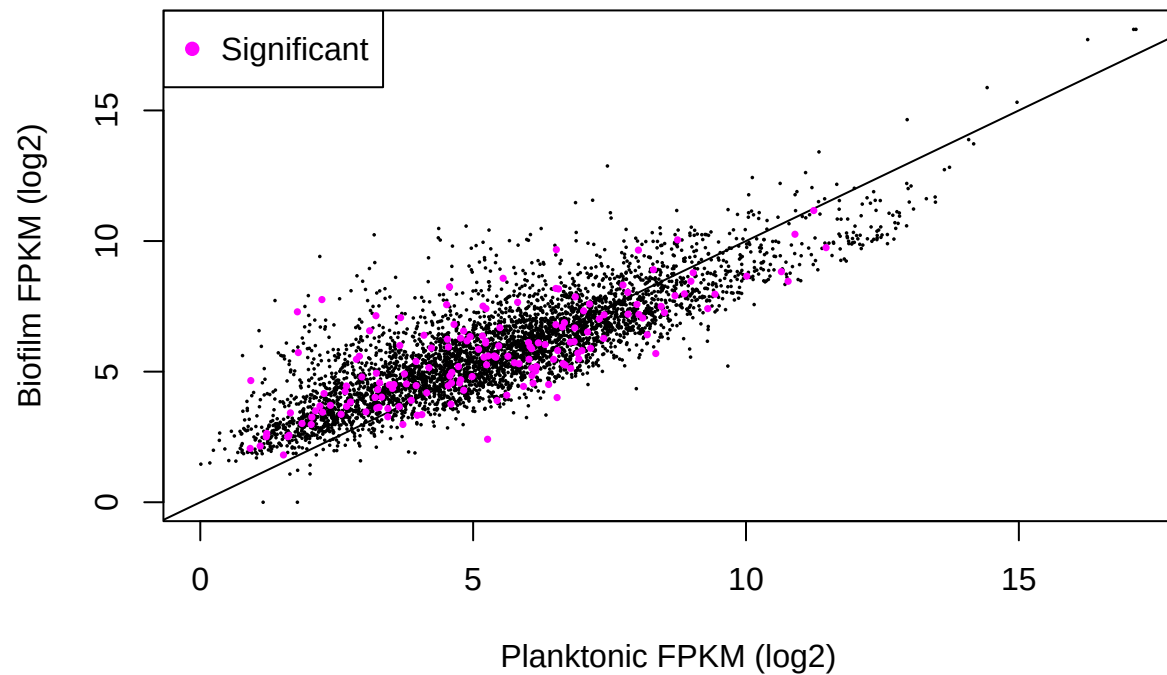


The figure above compares the $\log_2(\text{Fold change})$ for Planktonic vs Biofilm against the frequency at which it appears. There is horizontal symmetry in the fold changes, which is good because asymmetry indicates bias.

Plot total gene expression highlighting differentially expressed genes

```
gene_expression[, "plank"] = apply(gene_expression[, c(1:2)], 1, mean)
gene_expression[, "biofilm"] = apply(gene_expression[, c(3:4)], 1, mean)
x = log2(gene_expression[, "plank"] + min_nonzero)
y = log2(gene_expression[, "biofilm"] + min_nonzero)
plot(x=x, y=y, pch=16, cex=0.25, xlab="Planktonic FPKM (log2)", ylab="Biofilm FPKM (log2)",
     main="Planktonic vs Biofilm FPKMs")
abline(a=0, b=1)
xsig=x[sig]
ysig=y[sig]
points(x=xsig, y=ysig, col="magenta", pch=16, cex=0.5)
legend("topleft", "Significant", col="magenta", pch=16)
```

Planktonic vs Biofilm FPKMs



make a table of FPKM values

```
fpkm = texpr(bg_filt, meas="FPKM")
```

choose a gene to determine individual expression (pick a different number than I did)

```
ballgown::transcriptNames(bg_filt)[4]
```

```
##           4  
## "gene-PA0004"
```

```
ballgown::geneNames(bg_filt)[4]
```

```
##           4  
## "gyrB"
```

transform to log2

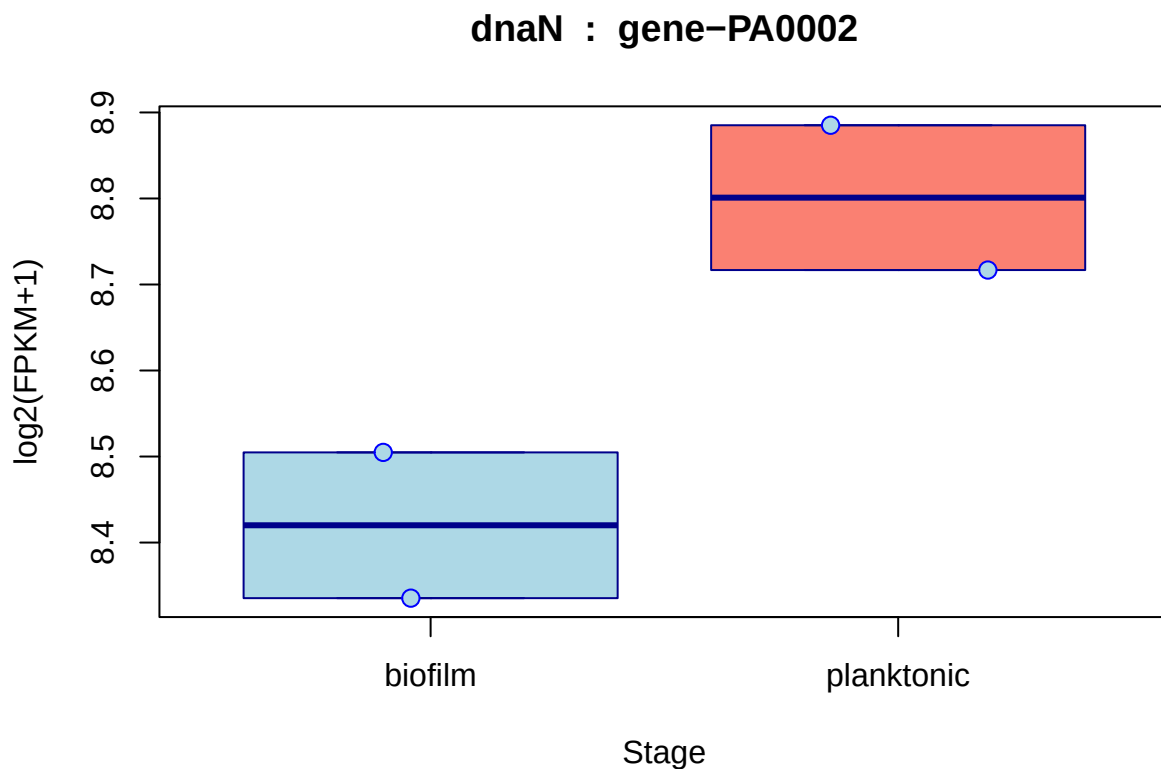
```
transformed_fpkm <- log2(fpkm[2, ] + 1)
```

make sure values are properly coded as numbers

```
numeric_stages <- as.numeric(factor(pheno_data$stage))  
jittered_stages <- jitter(numeric_stages)
```

plot expression of individual gene

```
boxplot(transformed_fpk ~ pheno_data$stage,  
  main=paste(ballgown::geneNames(bg_filt)[2], ' : ', ballgown::transcriptNames(bg_filt)[2]),  
  xlab="Stage",  
  ylab="log2(FPKM+1)",  
  col=c("lightblue", "salmon"),  
  border="darkblue")  
  
points(transformed_fpk ~ jittered_stages,  
  pch=21, col="blue", bg="lightblue", cex=1.2)
```



The figure above is comparing the biofilm and planktonic stage data against the $\log_2(\text{FPKM}+1)$ data.